

## Assignment 3

March 12, 2026

### Question 1: Mobile C.A.R.E. Foundation (MCF) Case [60 Points]

Recall the *Mobile C.A.R.E. Foundation (MCF)* case briefly discussed in class (see the IP2 slides).

MCF's mission is to control asthma in schoolchildren from low-income families in the greater Chicago area through the use of mobile asthma vans that visit elementary schools on weekdays. Their operations involve complex resource-allocation decisions aimed at maximizing service effectiveness.

Recently, MCF has faced a funding shortfall and needed to decrease the number of vans they operate. As a result, management must review performance metrics and implement improvement initiatives, including optimizing van scheduling and resource allocation across the 19 schools they serve, taking into account patient severity levels. You are provided with the following datasets (Table 1):

| Dataset                    | Description                                  |
|----------------------------|----------------------------------------------|
| School Locations           | (X, Y) coordinates for each school           |
| Case Patient List          | List of patients associated with each school |
| Case Patient Severity      | Severity level of each patient               |
| Case School K Appointments | Appointment details for school $K$           |

Table 1: Datasets provided by MCF and their descriptions.

Currently, MCF operates two vans, *Van 0* and *Van 1*. Your task is to determine:

- Which schools should be served by each van,
- How many appointments each van should provide, and
- Which patients will be scheduled for the next month.

Assume that vans operate only on weekdays and the data that was shared is recent.

1. **Exploratory Data Analysis (EDA): (5 points)** Perform basic EDA to estimate key parameters for your model (e.g., number of patients by severity level at each school, and van capacities inferred from historical data).
2. **Assumptions and Data Needs: (10 points)**
  - List all assumptions made while developing your model.
  - Discuss any additional data you wish you had to improve model validity and realism.
3. **Modeling: (20 points)** Using these insights, formulate a monthly scheduling model for both vans. Present your formulation in an **algebraic closed form**, clearly defining all decision variables, parameters, objective and constraints.
4. **Results: (20 points)** Solve the problem to optimality and provide a clear and interpretable schedule for both vans and their assigned patients.
5. **Objective Selection: (5 points)** Discuss how you have selected your current objective and how that could be modeled differently? What are the strengths and weaknesses of different approaches.

*Keep in mind that this is an open-ended question and we wish to see your approach to modeling! As long as your assumptions match with your modeling approach, you are good!*

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### Question 2: Branch and Bound [40 Points]

#### 2.A. (10 points)

Consider the following optimization problem featuring non-negative general integer variables:

$$\begin{aligned} \max \quad & z = 4x_1 - x_2 \\ \text{subject to} \quad & 7x_1 - 2x_2 \leq 14 \\ & x_2 \leq 3 \\ & 2x_1 - 2x_2 \leq 3 \\ & x_j \in \mathbb{Z}_+^2. \end{aligned}$$

Draw the feasible region of the integer program (possible as there are just two variables).

#### 2.B. (30 points)

Consider the IP in Question 2A and solve the problem using branch-and-bound. Draw out the tree, as well as the changes to the feasible region. Use the following criterion for node selection. For the first branch, choose the left-most branch. After this, always choose the right-most branch (if one exists), and prefer higher branches to lower branches.